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ITAI 1371 Introduction to Machine Learning

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Reflection Journal

This lab helped me to gain a better understanding of how pandas, NumPy, and matplotlib are used in machine learning and data analysis. Before completing the exercises, I knew these tools were important, but I did not fully understand how they worked together. Working through the notebook helped me see how data is loaded, analyzed, and visualized before it is used in a machine learning model.

One of the most valuable parts of the lab was working with pandas data frames. At first, viewing specific data was a little confusing, but after completing the lab, I became more comfortable with navigating the dataset. It reminded me of working with spreadsheets, but with much more flexibility. I also learned how useful summary statistics can be for understanding a dataset before performing deeper analysis.

The NumPy exercises helped me understand how numerical calculations can be performed efficiently on entire arrays of data. Seeing calculations such as averages and standard deviations completed with only a few lines of code showed me why NumPy is widely used in data science and machine learning.

I also enjoyed creating visualizations with matplotlib. The scatter plot of the Iris dataset made it easier to see patterns and differences between the flower species. This helped me understand that visualizations are not just for presentation purposes but are useful tools for exploring and understanding data.

Overall, this lab increased my confidence in using fundamental machine learning tools. I learned that data preparation, analysis, and visualization are important steps that help reveal patterns and improve understanding before building machine learning models. While I still need more practice with some pandas operations, I feel much more comfortable working with data than I did before completing this lab.

Reflective Journal: Module 2.

Nigel Bontiff

This lab was a good first step into actually using some of the tools that show up in machine learning. Before doing it, I had heard of things like pandas, NumPy, matplotlib, and scikit-learn, but I did not really understand how they worked together. I mostly thought machine learning was just about training a model and generating predictions. As I work through this lab, I can see that much of the process starts before the model part even happens. You first have to load the data, organize it, look at it, graph it, and understand what the data is showing. One thing I liked about this lab was that it used the Iris dataset, which was simple enough to follow but still showed real machine learning ideas. The dataset had 150 samples, four measurement features, and three flower species. Seeing that structure helped me understand what people mean when they talk about features and classes. The measurements like sepal length, sepal width, petal length, and petal width are the features, and the flower species are the classes. That made the terms feel a lot less abstract. The biggest learning moment for me was probably creating the scatter plot. Looking at the table of numbers was useful, but the graph made the patterns way easier to see. Once the points were plotted by sepal length and sepal width, it became clear that setosa was easier to separate from the other two species. At the same %me, versicolor and virginica overlapped more in the middle of the graph. That helped me understand why visualization matters. It is not just something extra to make the assignment look better. It actually helps you notice patterns that would be harder to catch by only reading the data. I also started to see how each tool has its own role. Pandas made the dataset easier to work with by putting it into a Data Frame. NumPy was useful for calculations like finding the mean and standard deviation of sepal length. Matplotlib was used to create the scatter plot and bar chart. Scikit-learn provided the Iris dataset. At first, it felt like a lot of different tools at once, but once I understood the purpose of each one, the workflow made more sense. Another thing I learned was that basic statistics are still important in machine learning. The mean sepal length gave me a quick idea of the average size in the dataset, while the standard deviation showed how spread out the values were. I used to think statistics were separate from machine learning, but now I can see that they are connected. A model is basically trying to learn from patterns in the data, so it makes sense that we should understand the numbers before jumping into modeling. Nigel Bonti+ 2 The species count bar chart was also helpful, even though it was simple. It showed that each species had 50 samples. That made me realize that checking the balance of a dataset matters. If one species had way more examples than the others, that could affect how a model learns later. Even though we did not train a model in this lab, I can already see how early data exploration can help prevent problems later on. One challenge for me was getting used to the syntax. Some of the code was easy to understand, like using df.head() to see the first few rows. Other parts took more action, especially grouping the data by species and looping through each species to plot the points. I had to remind myself that the goal is not to memorize every command right away. The goal is to understand what the code is doing and why it is useful. Overall, this lab helped me understand that machine learning starts with understanding the data. Before this assignment, I probably would have skipped straight to wanting to build a model. Now I see why data exploration is such an important first step. If I do not understand the data, then I cannot really understand what the model is learning from it. Going forward, I want to get better at reading graphs and asking better questions from them. For example, AKer seeing the overlap between versicolor and virginica, I would want to know whether petal length and petal width would separate the species better than sepal length and sepal width. I would also want to understand which features are most useful for classification. This assignment made the machine learning process feel more realistic to me. It showed me that tools like pandas, NumPy, matplotlib, and scikit-learn are not just random libraries to memorize. They each help with a different part of working with data. The main thing I took away is that good machine learning depends on good data understanding first.

Ved Dalwadi

In this journal entry, I will share my thoughts and experiences about the Module 2 lab exercise. In this course, I have learned to use Google Collab to run code and tools used in machine learning, such as different Python libraries that include Machine Learning datasets for me to experiment with. In this lab, I was engaged with Pandas and Matplotlib which showed me the prep that goes into Machine Learning datasets. The point of this reflection is to show what I have learned in this module and how I can use it in the future. In this lab, I used Google Collab to show the Iris dataset and how this dataset and other tools set the groundwork for machine learning. In this lab, I was shown how to use different tools of machine learning, such as Pandas, which is a very important part of the success of AI data training, as all machine learning algorithms need organized data to be able to function properly, which Pandas will provide for the AI. This was a surprise for me because I had just believed that the algorithm was the only important part of machine learning and did not consider the dataset used to train it is also important. Another important tool for machine learning is NumPy, which is a very fast and incredibly efficient calculator used for high speed machine learning datasets. I researched the last generation of Python calculators, Numeric, which was considered too slow for most AI neural networks, which need millions of matrix multiplications, which the previous Numeric could not produce, which is why NumPy was produced to keep up with the AI neural networks. These key tools helped me understand how the basic setup for machine learning works and how to do this in the future. When I first started the lab, I didn't understand what was happening because I didn't know Python, so I didn't understand the library's given or how it related to machine learning. However, as I continued to run the code more and look at it, I started to understand each part of the code. I slowly started to see that the dataset was given to Pandas to organize the dataset in rows and columns. Another tool used is matplotlib, which is important for showing the data, which helps data scientists determine if parts of the data are important or not. As I worked through the lab, I had a few questions that I had thought about. I first wondered how someone could build an algorithm to predict how a flower will grow based on the breed. As a human, I can see that multiple flowers of a certain breed end up in a cluster that is similar, but I don't understand how you can give this raw data to an AI and train it to be able to understand how to predict the growth of the breed. Another question I had while working through the lab was how Pandas could handle a very large, complex data set, as the iris data set is already very organized, so I wondered how it could handle this and if humans were needed to clean the complex dataset. This lab helped me grow as it showed me the prep work about things I only use and hear about and never interact with. This lab showed that machine learning is many different things, not just AI. Something that surprised me is the effort in the dataset, where I knew that AI used data sets to train based on, but I never knew that you just can't have any random piece of data collection, and it needs to be structured data for it to function. This lab helped me learn new things I had not yet learned, like the intense amount of matrix multiplication needed to help build an AI. When working through this lab, I now have a better understanding of how machine learning works and now want to learn more about how machine learning and AI systems work and are made today